

STA 2101: Statistics & Probability

Lecture 05: Charts and Graphs Continued

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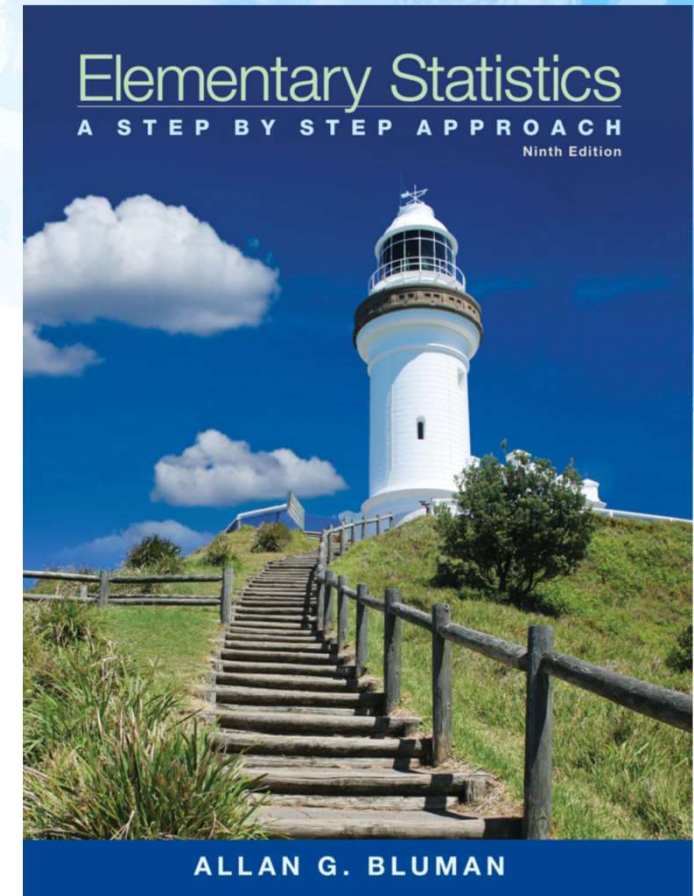
Class Code
sdb2r4mu



Fall 2025

Reference Book

- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution



Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today's Topics

- Last Class +
- Relative Frequency Graphs
- Charts

Types of Representation

Last Class
(Lecture 4)

- Table
- Graphs

Cumulative frequency table of the ages of survey participants

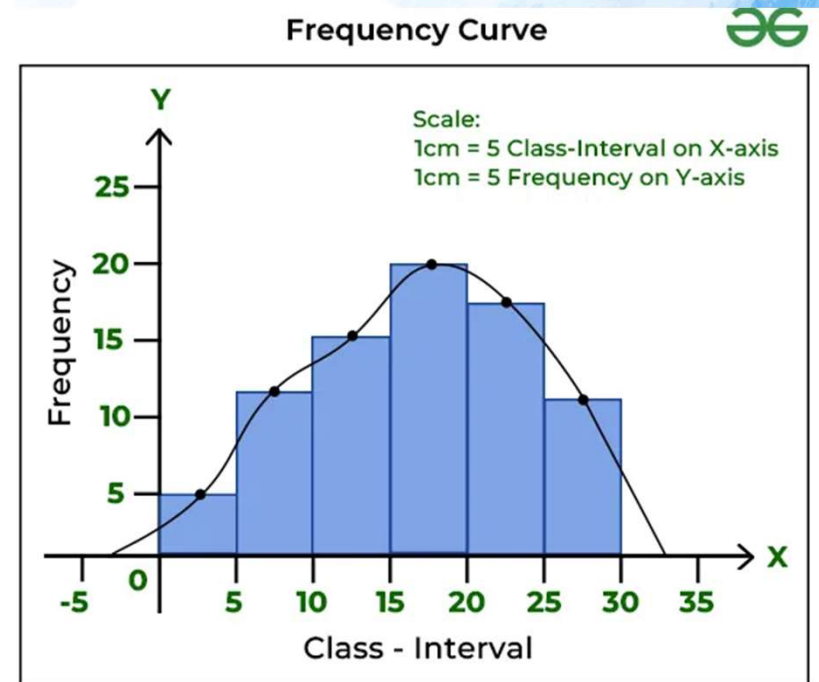
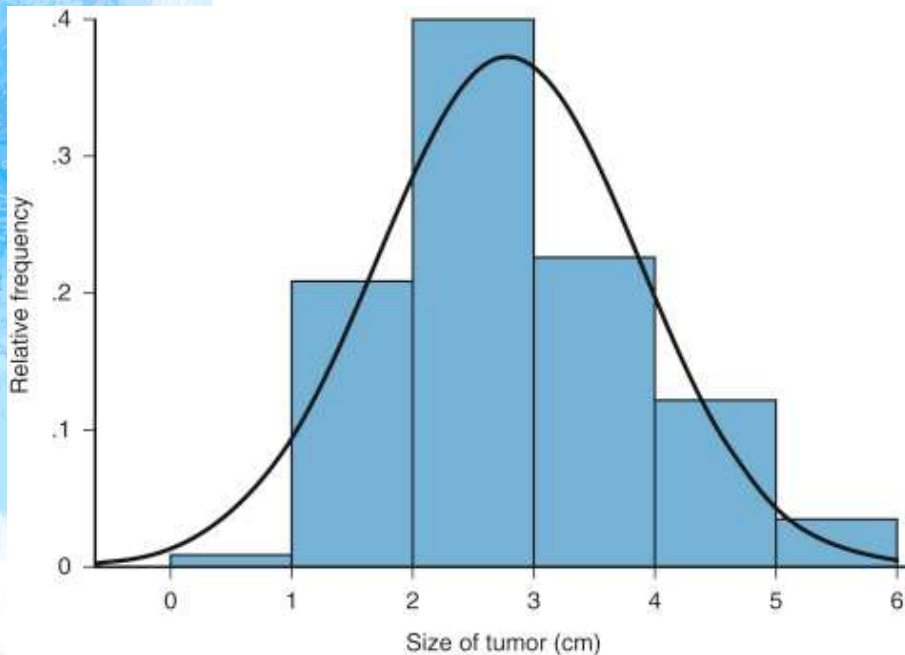
Age, a (years)	Frequency	Cumulative frequency	Cumulative relative frequency
$19 \leq a < 29$	4	4	$4 / 20 = .2$
$29 \leq a < 39$	9	$9 + 4 = 13$	.65
$39 \leq a < 49$	3	$9 + 4 + 3 = 16$	.8
$49 \leq a < 59$	3	19	.95
$59 \leq a < 69$	1	20	1



Last Class (Lecture 4)

Types of Representation

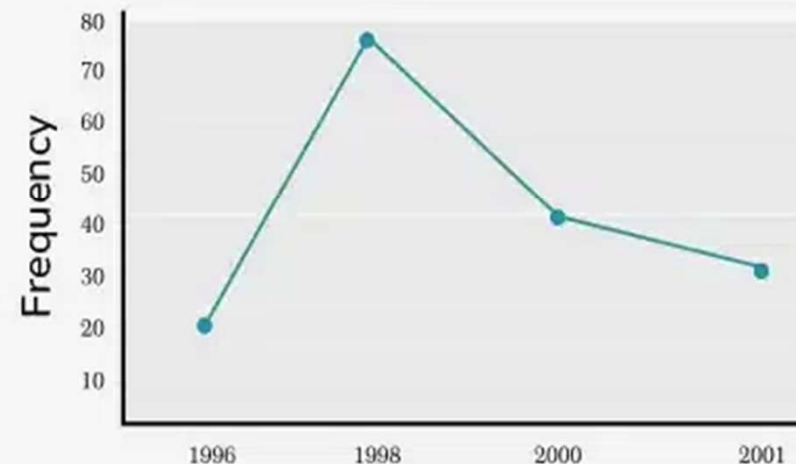
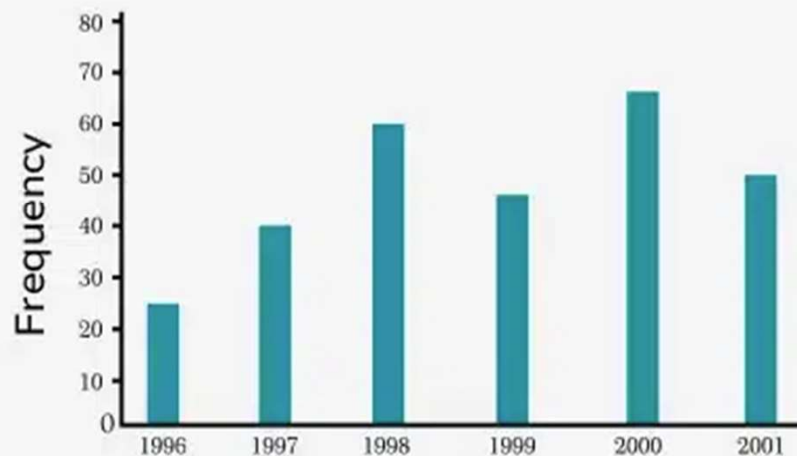
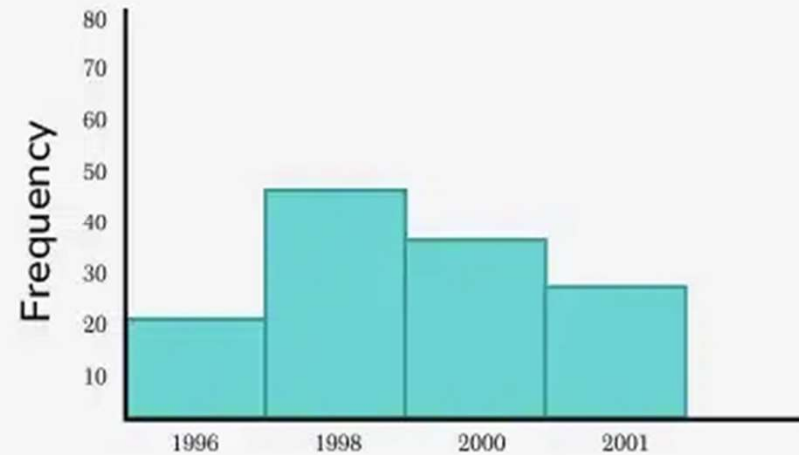
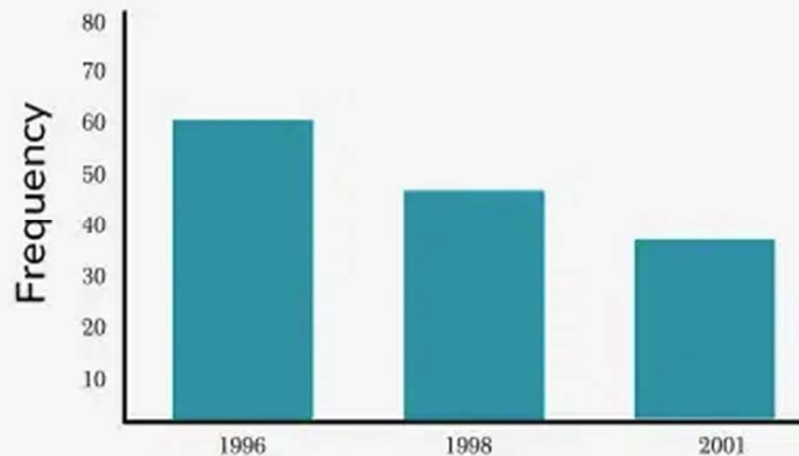
- Table
- Graphs



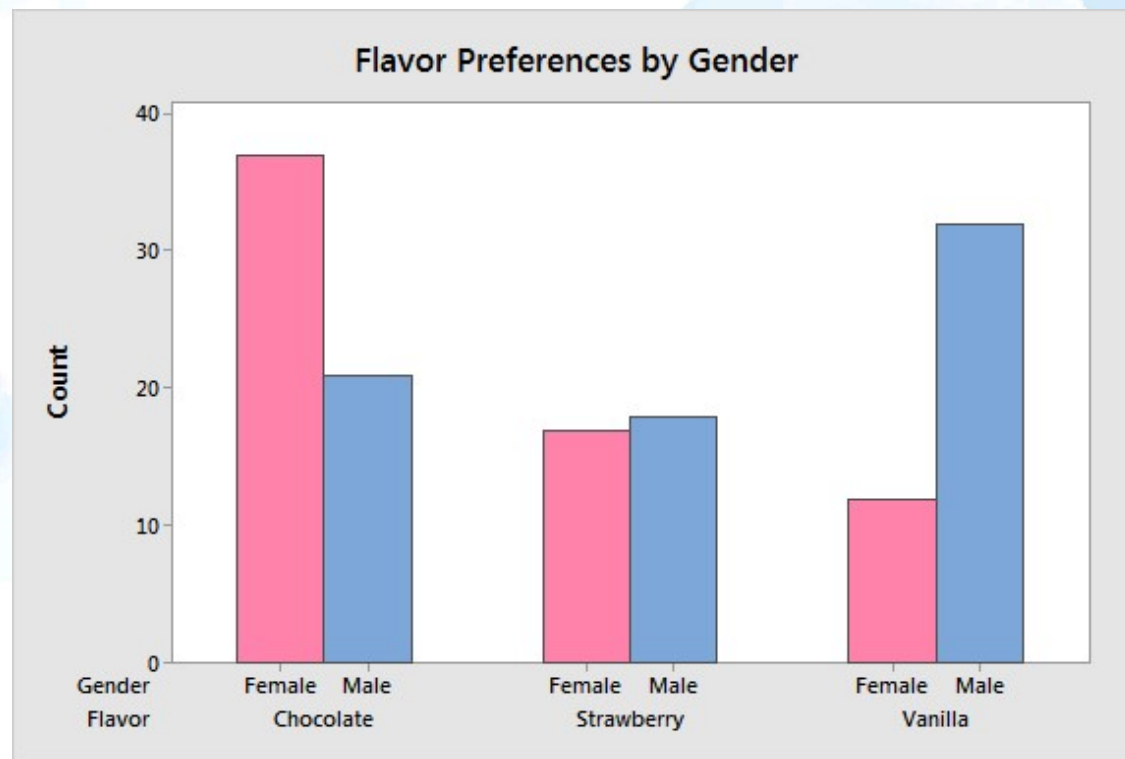
FREQUENCY DISTRIBUTION

Frequency Distribution is a tool in statistics that helps us organize the data and also helps us reach meaningful conclusions.

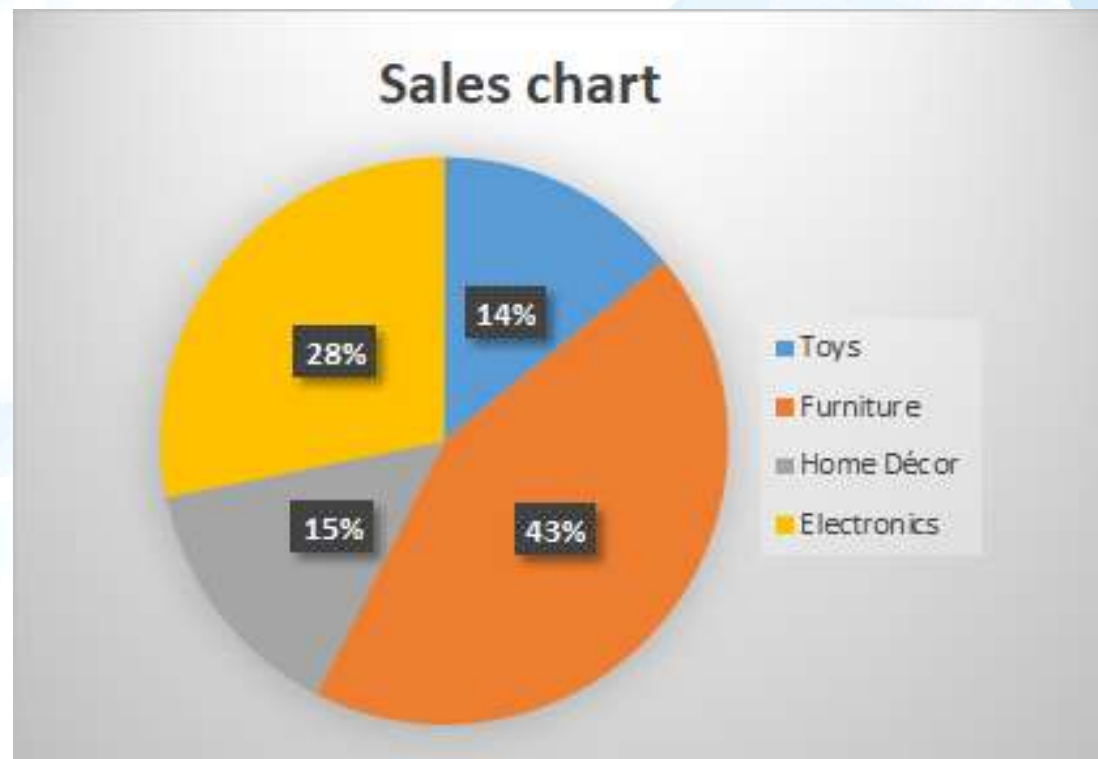
Last Class
(Lecture 4)



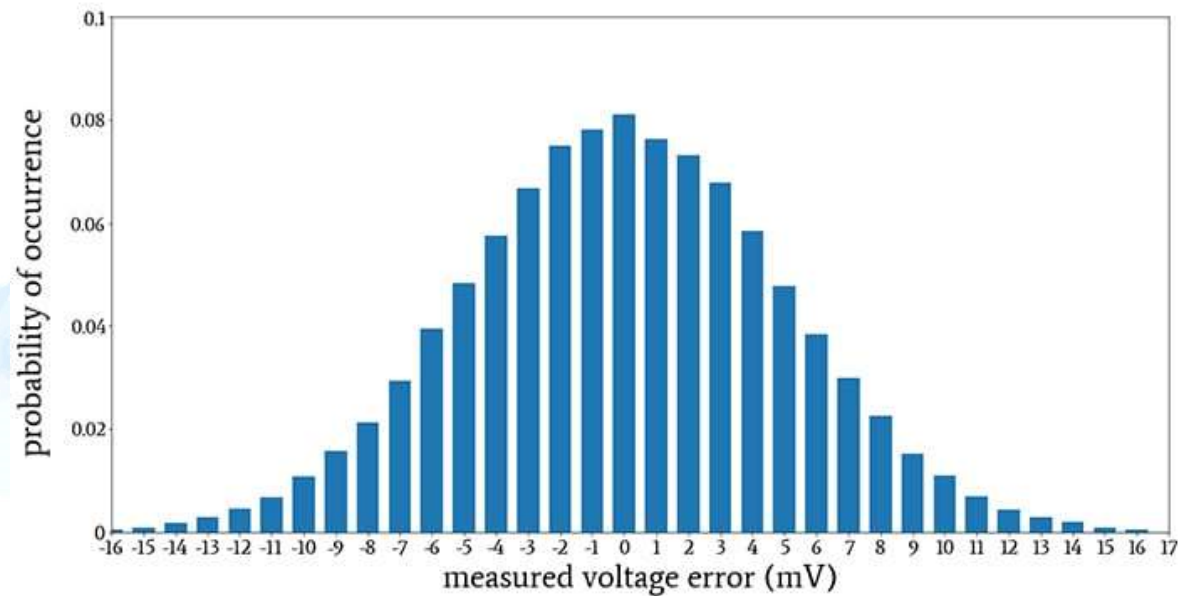
Bar Chart



Pie Chart



Histogram



Types of Histogram

Last Class
(Lecture 4)

Types of Histograms (by Shape of Distribution)

1. Uniform Distribution

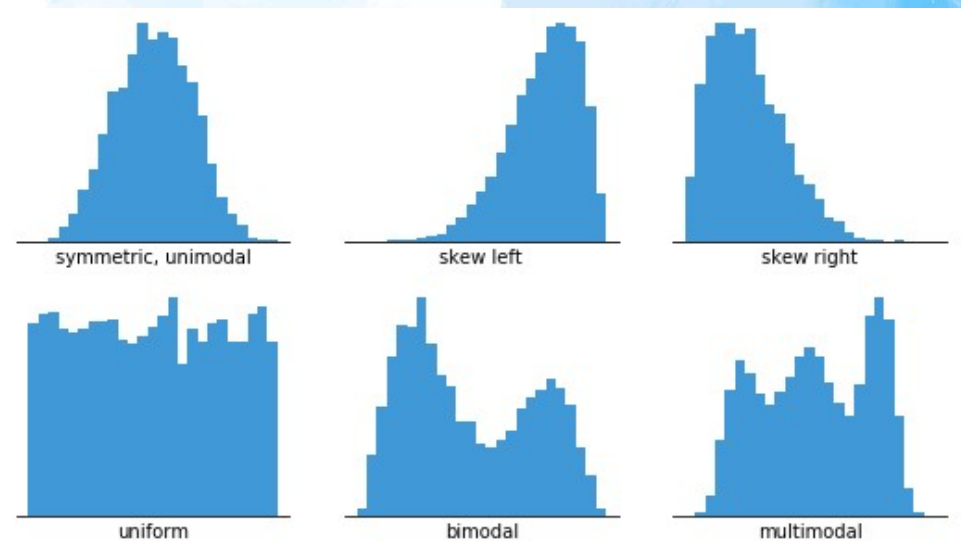
- All bars are roughly the same height.
- Data values are spread out evenly.
- **Example:** Rolling a fair die many times → each number (1–6) occurs equally often.

2. Symmetric (Bell-Shaped / Normal Distribution)

- The histogram is **balanced** around the center.
- Bars rise to a peak in the middle and taper off on both sides.
- **Example:** Heights of people, exam scores (in large groups).

3. Skewed Right (Positively Skewed)

- Tail (longer side) extends to the **right**.
- Most values are on the lower side, but a few large values pull the tail.
- **Example:** Income distribution (many earn average, few earn extremely high).



Types of Histogram

Last Class
(Lecture 4)

4. Skewed Left (Negatively Skewed)

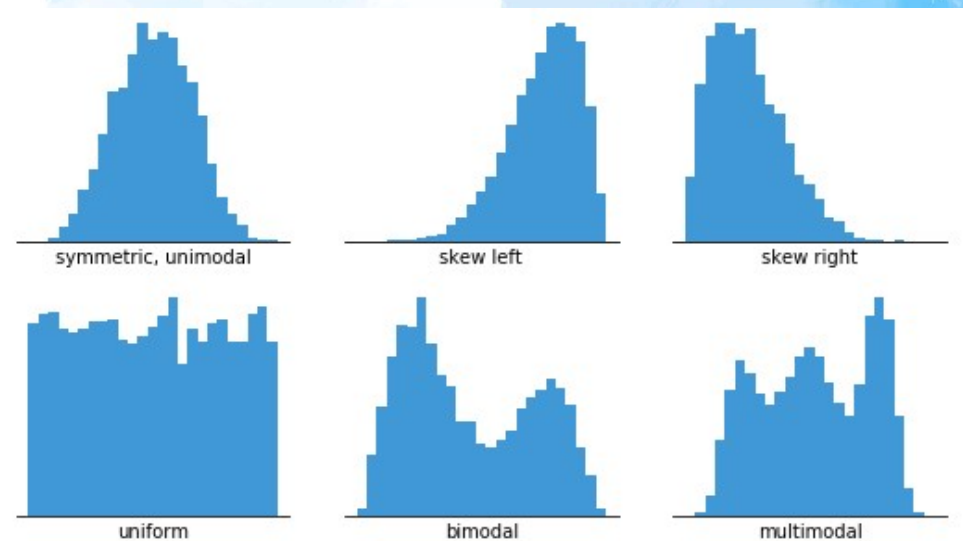
- Tail (longer side) extends to the **left**.
- Most values are on the higher side, but a few very small values pull the tail.
- **Example:** Age at retirement (most retire around 60–65, few retire much earlier).

5. Bimodal Distribution

- Two distinct peaks (modes).
- Suggests data comes from **two different groups**.
- **Example:** Test scores of a mixed class (strong group + weak group).

6. Multimodal Distribution

- More than two peaks.
- Indicates multiple subgroups in the data.
- **Example:** Daily traffic flow (morning, noon, evening peaks).



From the book

Last Class
(Lecture 4)

2-2 Histograms, Frequency Polygons, and Ogives

The Histogram

The **histogram** is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

EXAMPLE 2-4 Record High Temperatures

Construct a histogram to represent the data shown for the record high temperatures for each of the 50 states (see Example 2-2).

Class boundaries	Frequency
99.5–104.5	2
104.5–109.5	8
109.5–114.5	18
114.5–119.5	13
119.5–124.5	7
124.5–129.5	1
129.5–134.5	1

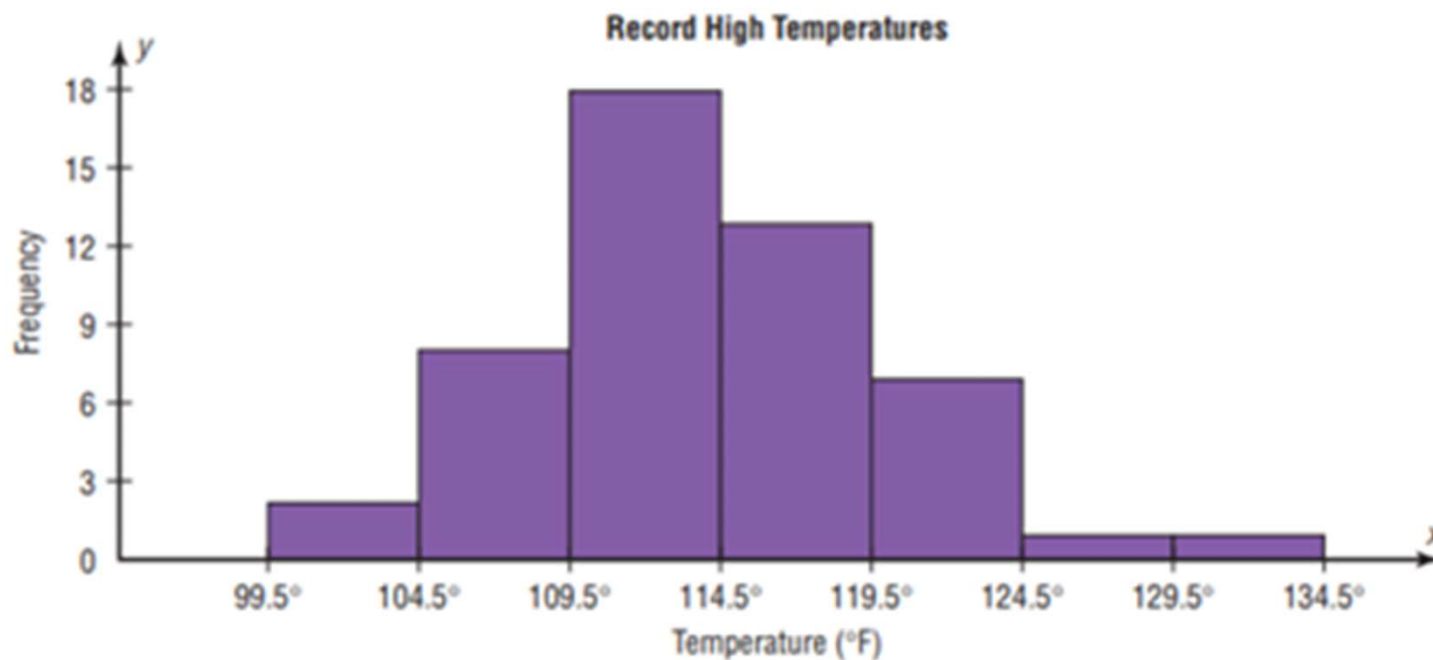
SOLUTION

- Step 1** Draw and label the x and y axes. The x axis is always the horizontal axis, and the y axis is always the vertical axis.
- Step 2** Represent the frequency on the y axis and the class boundaries on the x axis.
- Step 3** Using the frequencies as the heights, draw vertical bars for each class. See Figure 2-1.

From the book

Last Class
(Lecture 4)

FIGURE 2-1 Histogram for Example 2-4



As the histogram shows, the class with the greatest number of data values (18) is 109.5–114.5, followed by 13 for 114.5–119.5. The graph also has one peak with the data clustering around it.

From the book

The Frequency Polygon

Another way to represent the same data set is by using a frequency polygon.

The **frequency polygon** is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

EXAMPLE 2-5 Record High Temperatures

Using the frequency distribution given in Example 2-4, construct a frequency polygon.

SOLUTION

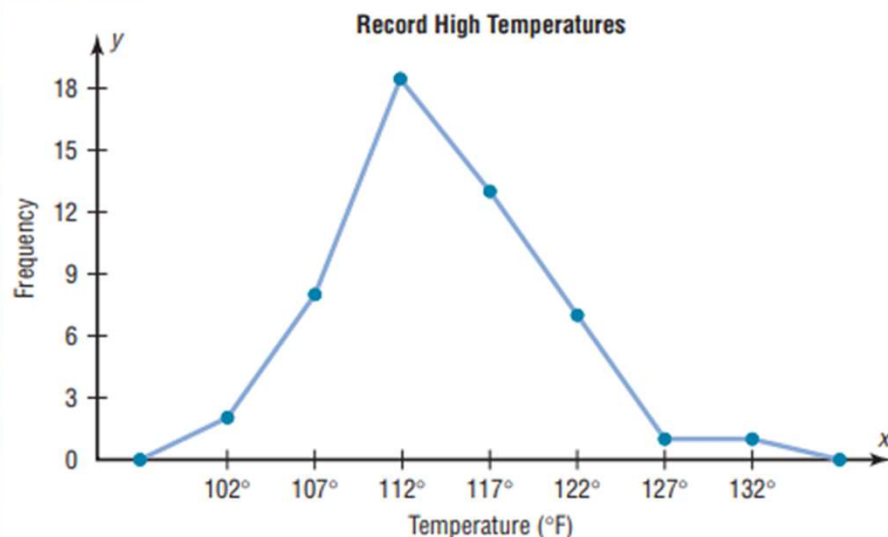
Step 1 Find the midpoints of each class. Recall that midpoints are found by adding the upper and lower boundaries and dividing by 2:

$$\frac{99.5 + 104.5}{2} = 102 \quad \frac{104.5 + 109.5}{2} = 107$$

and so on. The midpoints are

Class boundaries	Midpoints	Frequency
99.5–104.5	102	2
104.5–109.5	107	8
109.5–114.5	112	18
114.5–119.5	117	13
119.5–124.5	122	7
124.5–129.5	127	1
129.5–134.5	132	1

- Step 2** Draw the x and y axes. Label the x axis with the midpoint of each class, and then use a suitable scale on the y axis for the frequencies.
- Step 3** Using the midpoints for the x values and the frequencies as the y values, plot the points.
- Step 4** Connect adjacent points with line segments. Draw a line back to the x axis at the beginning and end of the graph, at the same distance that the previous and next midpoints would be located, as shown in Figure 2-2.



Last Class
(Lecture 4)

From the book

The Ogive

The third type of graph that can be used represents the cumulative frequencies for the classes. This type of graph is called the *cumulative frequency graph*, or *ogive*. The **cumulative frequency** is the sum of the frequencies accumulated up to the upper boundary of a class in the distribution.

The **ogive** is a graph that represents the cumulative frequencies for the classes in a frequency distribution.

EXAMPLE 2-6 Record High Temperatures

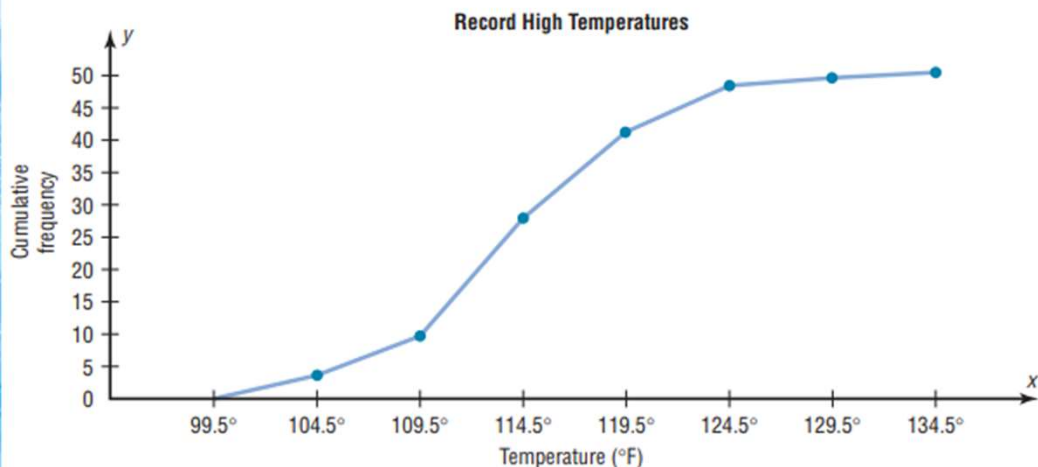
Construct an ogive for the frequency distribution described in Example 2-4.

SOLUTION

Step 1 Find the cumulative frequency for each class.

	Cumulative frequency
Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50

- Step 2** Draw the x and y axes. Label the x axis with the class boundaries. Use an appropriate scale for the y axis to represent the cumulative frequencies. (Depending on the numbers in the cumulative frequency columns, scales such as 0, 1, 2, 3, . . . , or 5, 10, 15, 20, . . . , or 1000, 2000, 3000, . . . can be used. Do *not* label the y axis with the numbers in the cumulative frequency column.) In this example, a scale of 0, 5, 10, 15, . . . will be used.
- Step 3** Plot the cumulative frequency at each upper class boundary, as shown in Figure 2-3. Upper boundaries are used since the cumulative frequencies represent the number of data values accumulated up to the upper boundary of each class.
- Step 4** Starting with the first upper class boundary, 104.5, connect adjacent points with line segments, as shown in Figure 2-4. Then extend the graph to the first lower class boundary, 99.5, on the x axis.



Last Class
(Lecture 4)

Lecture 5

Relative Frequency Graphs + Charts

[From Book]

Relative Frequency Graphs

The histogram, the frequency polygon, and the ogive shown previously were constructed by using frequencies in terms of the raw data. These distributions can be converted to distributions using *proportions* instead of raw data as frequencies. These types of graphs are called **relative frequency graphs**.

EXAMPLE 2-7 Miles Run per Week

Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the miles that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	Total 20

SOLUTION

Step 1 Convert each frequency to a proportion or relative frequency by dividing the frequency for each class by the total number of observations.

For class 5.5–10.5, the relative frequency is $\frac{1}{20} = 0.05$; for class 10.5–15.5, the relative frequency is $\frac{2}{20} = 0.10$; for class 15.5–20.5, the relative frequency is $\frac{3}{20} = 0.15$; and so on.

Place these values in the column labeled Relative frequency.

Class boundaries	Midpoints	Relative frequency
5.5–10.5	8	0.05
10.5–15.5	13	0.10
15.5–20.5	18	0.15
20.5–25.5	23	0.25
25.5–30.5	28	0.20
30.5–35.5	33	0.15
35.5–40.5	38	0.10
		<u>Total 1.00</u>

SOLUTION

Step 1 Convert each frequency to a proportion or relative frequency by dividing the frequency for each class by the total number of observations.

For class 5.5–10.5, the relative frequency is $\frac{1}{20} = 0.05$; for class 10.5–15.5, the relative frequency is $\frac{2}{20} = 0.10$; for class 15.5–20.5, the relative frequency is $\frac{3}{20} = 0.15$; and so on.

Place these values in the column labeled Relative frequency.

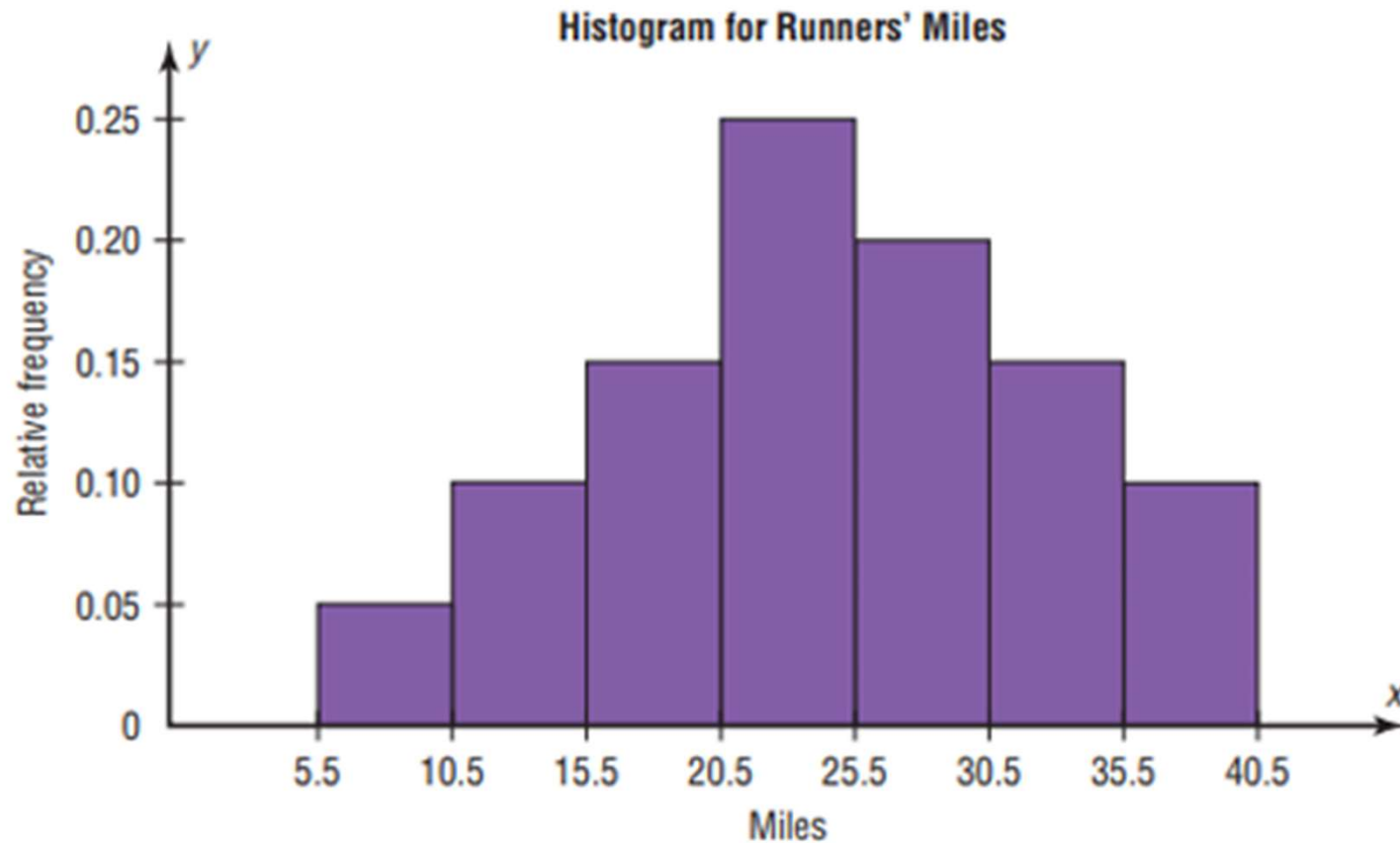
Class boundaries	Midpoints	Relative frequency
5.5–10.5	8	0.05
10.5–15.5	13	0.10
15.5–20.5	18	0.15
20.5–25.5	23	0.25
25.5–30.5	28	0.20
30.5–35.5	33	0.15
35.5–40.5	38	0.10
		<u>Total 1.00</u>

Step 2 Find the cumulative relative frequencies. To do this, add the frequency in each class to the total frequency of the preceding class. In this case, $0 + 0.05 = 0.05$, $0.05 + 0.10 = 0.15$, $0.15 + 0.15 = 0.30$, $0.30 + 0.25 = 0.55$, etc. Place these values in the column labeled Cumulative relative frequency.

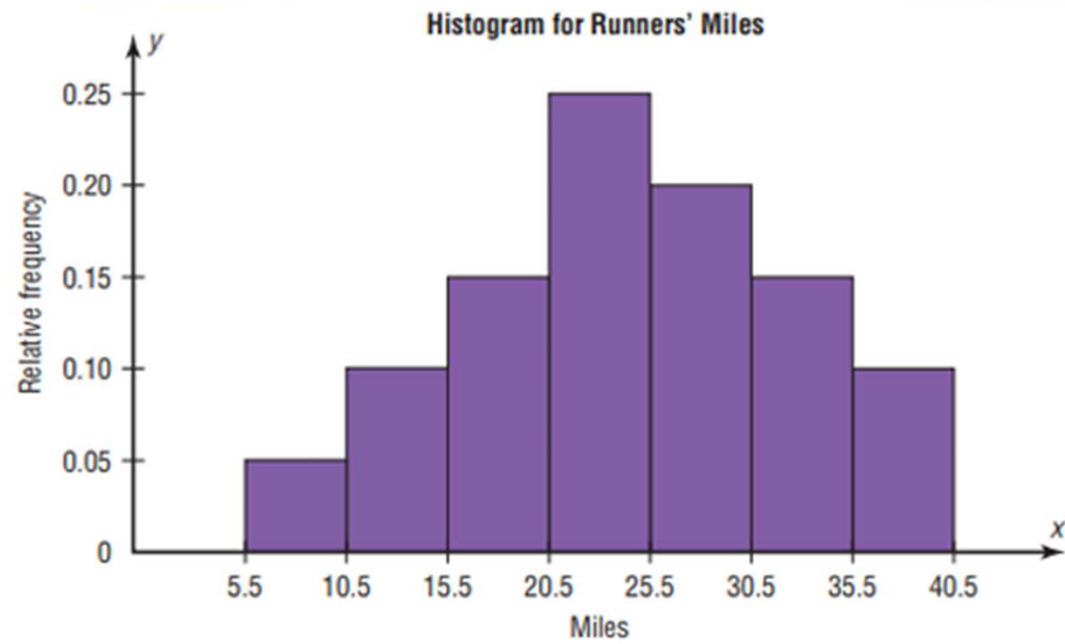
An alternative method would be to find the cumulative frequencies and then convert each one to a relative frequency.

	Cumulative frequency	Cumulative relative frequency
Less than 5.5	0	0.00
Less than 10.5	1	0.05
Less than 15.5	3	0.15
Less than 20.5	6	0.30
Less than 25.5	11	0.55
Less than 30.5	15	0.75
Less than 35.5	18	0.90
Less than 40.5	20	1.00

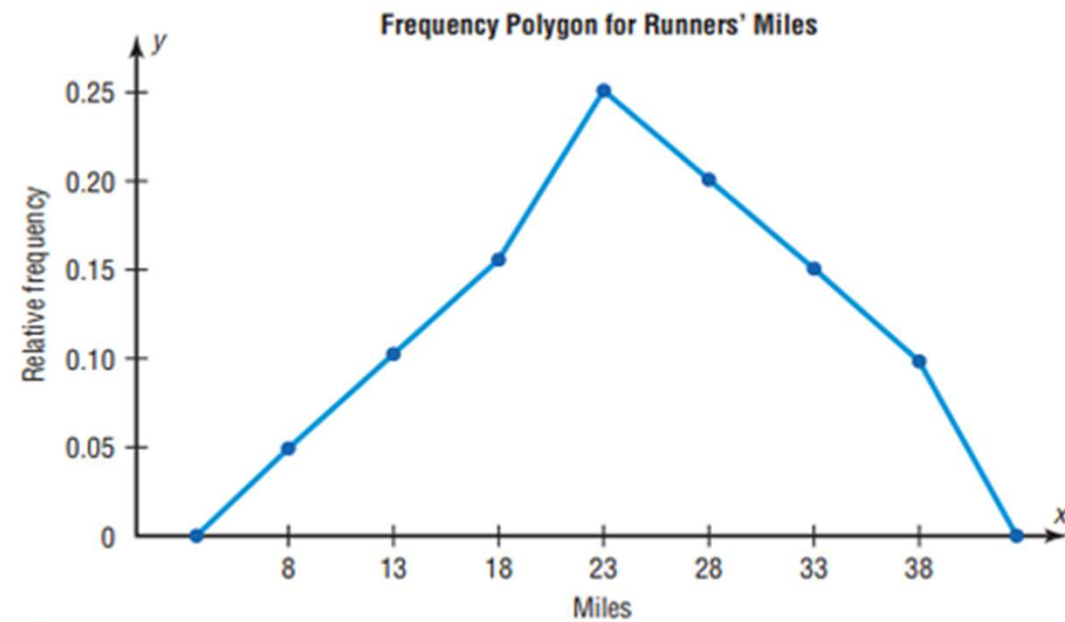
Step 3 Draw each graph as shown in Figure 2–6. For the histogram and ogive, use the class boundaries along the x axis. For the frequency polygon, use the midpoints on the x axis. The scale on the y axis uses proportions.



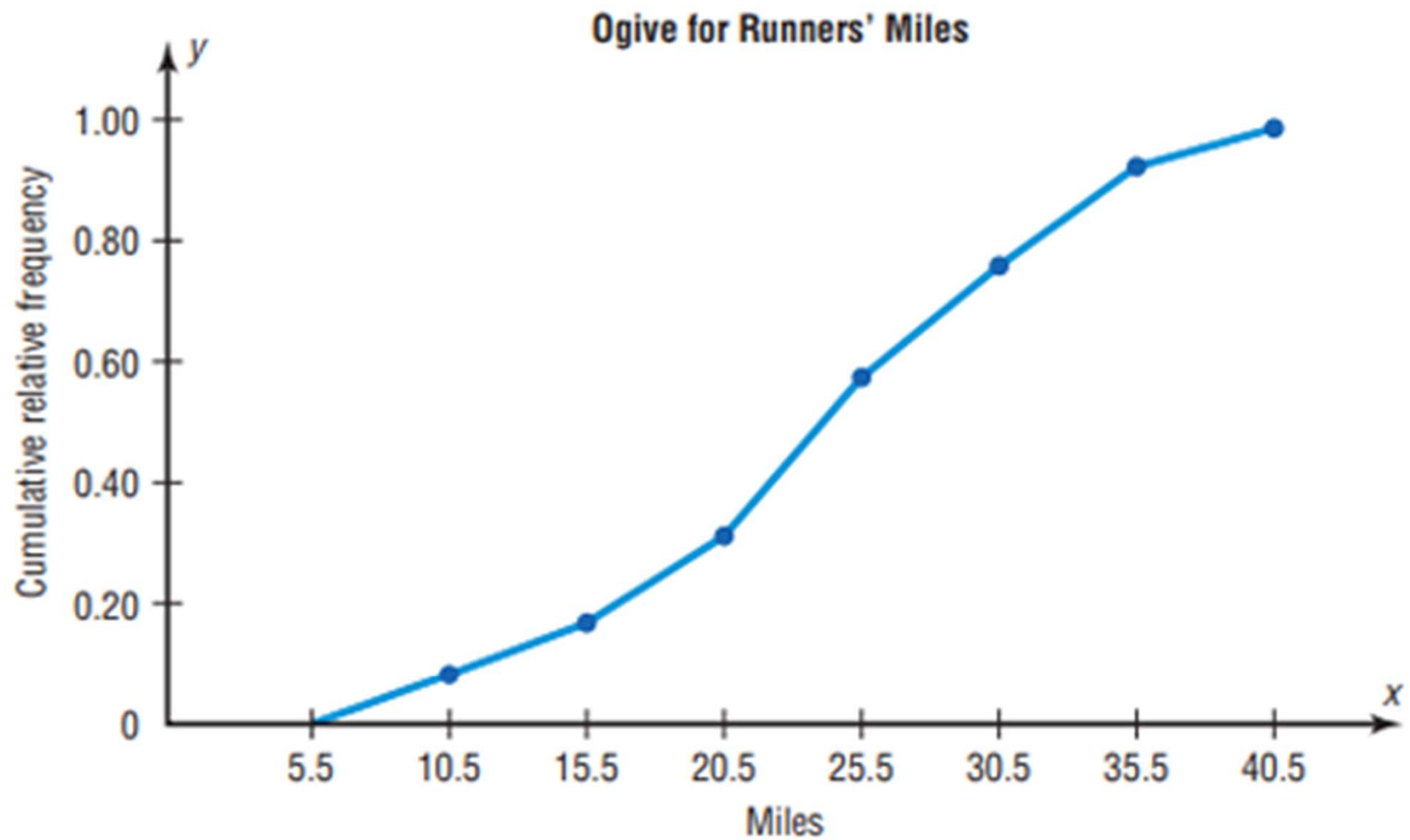
(a) Histogram



(a) Histogram

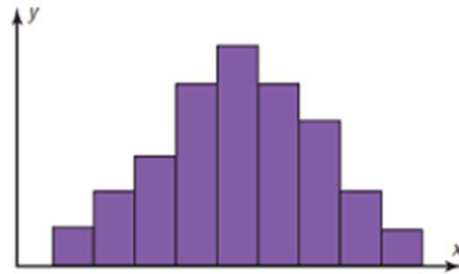


(b) Frequency polygon

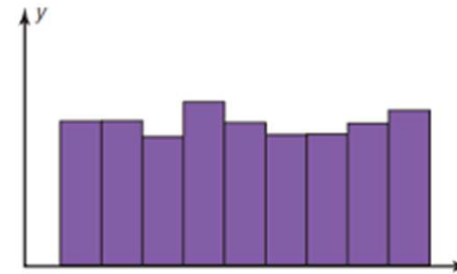


(c) Ogive

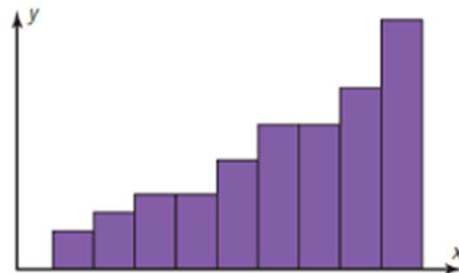
Distribution Shapes



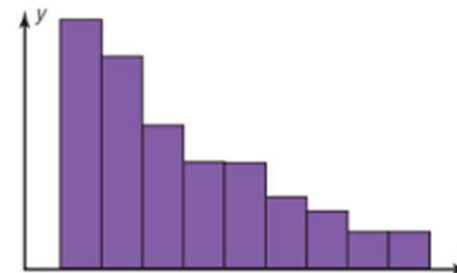
(a) Bell-shaped



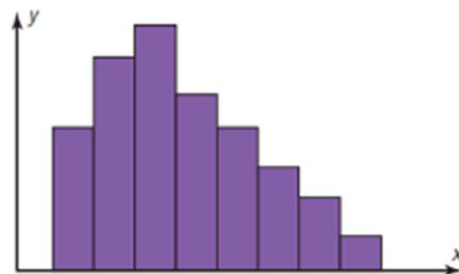
(b) Uniform



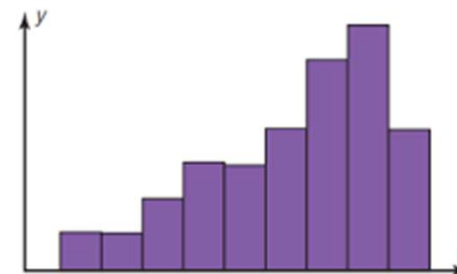
(c) J-shaped



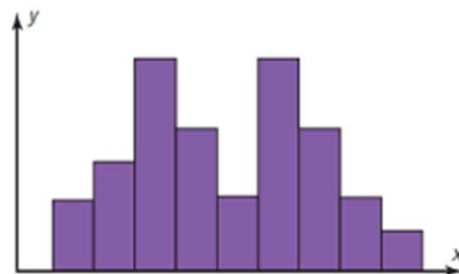
(d) Reverse J-shaped



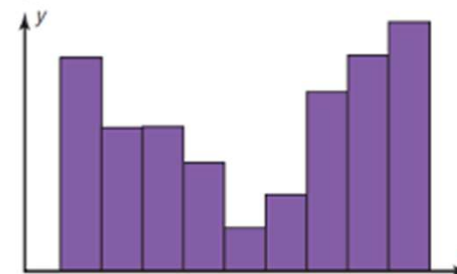
(e) Right-skewed



(f) Left-skewed



(g) Bimodal



(h) U-shaped



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Thank You

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